

Convergence Investigation Report — case-a

Generated: 2026-09-29T12:41:40.961536Z

Investigation Identities

Original: **caseA_researcher**
Candidate: **caseA_unrelated**

Linkage Strength-of-Evidence

16.0/100 — LOW

This score represents strength of evidence, not probability of guilt or identity certainty.

Signal Breakdown

Signal	Similarity	Contribution	Status	Evidence ID
temporal	0.138	+2.95	INSUFFICIENT	E-caseA_researcher-caseA_unrelated-temporal
account	0.000	+0.00	INSUFFICIENT	E-caseA_researcher-caseA_unrelated-account
infrastructure	0.050	+0.68	INSUFFICIENT	E-caseA_researcher-caseA_unrelated-infrastructure
activity	0.639	+12.33	SUPPORTING	E-caseA_researcher-caseA_unrelated-activity
interaction_network	0.000	+0.00	INSUFFICIENT	E-caseA_researcher-caseA_unrelated-interaction_network
linguistic	0.000	+0.00	INSUFFICIENT	E-caseA_researcher-caseA_unrelated-linguistic

Supporting Evidence

activity: {'identity_a_actions': {'message': 3, 'post': 2, 'upload': 5, 'search': 3, 'withdraw': 1, 'browse': 2}, 'identity_b_actions': {'withdraw': 3, 'upload': 2, 'search': 2}, 'strongest_matching_behaviors': ['search', 'upload', 'withdraw'], 'similarity': 0.639}

Contradicting Evidence

None identified.

Insufficient Evidence

temporal: {'matching_activity_hours': [0, 1, 4, 23], 'matching_activity_windows': ['00:00–02:00', '04:00–05:00', '23:00–24:00'], 'observation_period_days': 14, 'similarity': 0.138}

account: {'creation_gap_days': 75.0, 'posting_interval_diff_minutes': 220.0, 'creation_decay_days': 60, 'posting_decay_minutes': 120, 'similarity': 0.0}

infrastructure: {'identity_a_infra': 'infra_A_PRIVACY', 'identity_b_infra': 'infra_B_UNRELATED', 'matched_components': [], 'match_ratio': 0.0, 'match_type': 'none', 'identity_a_components': {'tls': 'tls_2', 'service': 'svc_7', 'certificate': 'cert_10', 'server': 'srv_6', 'asn': 'asn_2'}, 'identity_b_components': {'tls': 'tls_UNIQUE_B', 'service': 'svc_UNIQUE_B', 'certificate': 'cert_UNIQUE_B', 'server': 'srv_UNIQUE_B', 'asn': 'asn_UNIQUE_B'}, 'similarity': 0.05, 'ps26151_infrastructure_evidence': {'ssl_certificates': [], 'certificate_domain_links': [], 'clearnet_domains': [], 'service_banners': ['svc_7'], 'exposed_server_status': [], 'descriptor_inconsistencies': []}, 'observations': ['Shared service/default banner(s).'], 'note': 'Infrastructure score uses TLS/service/certificate/server/ASN component overlap; PS-specific clearnet/certificate/server-status/descriptor objects are explicitly represented as corroborating evidence.'}

interaction_network: {'identity_a_interaction_vector': [10, 0, 0, 0, 0], 'identity_b_interaction_vector': [0, 10, 0, 0, 0], 'shared_counterparties': [], 'shared_counterparty_count': 0, 'similarity': 0.0, 'note': 'Behavioral interaction patterns are attribution evidence; privacy/anonymity tools are excluded from this signal.'}

linguistic: {'similarity': 0.0, 'features': ['sentence length', 'punctuation pattern', 'function-word usage', 'vocabulary patterns', 'n-gram distribution'], 'representation': 'Synthetic stylometric representation', 'note': 'Prototype stylometry is synthetic; the similarity computation itself is cosine similarity.'}

Threat-Actor Intelligence Layer

Shared objects: **1**
Handles: []

PGP keys: []
Wallets: []
Trust links: []
Marketplaces: []
Cleartext domains: []
SSL certificates: []
Certificate ↔ domain links: []
Service/default banners: ['svc_7']
Exposed server-status: []
Descriptor inconsistencies: []
Hidden-service identifiers: []
Persona history overlap: []
This layer is corroborating context and is not a seventh score signal.

Privacy / Anonymity Context — NOT Attribution Evidence

Original: {'tor': True, 'vpn': True, 'cryptocurrency': True, 'encrypted_messaging': True}
Candidate: {'tor': False, 'vpn': False, 'cryptocurrency': True, 'encrypted_messaging': True}
These indicators do not contribute to the attribution score.

Methodology

Temporal cosine similarity; configurable account decay; infrastructure component overlap; activity cosine similarity; interaction-network similarity; synthetic stylometry; weighted evidence fusion; separate contradiction penalty; confidence bands.

Evidence Integrity

E-caseA_researcher — SHA-256: ee57eba604a599da80d37ffd97d88a66cd94e59203016d91e4aa443df40a9ea7

E-caseA_unrelated — SHA-256: faa86e7cc804ca6256d8e6f1331fc23cb2b332f5d96a9ac5ddf31b74ec57abbe

Limitations

Controlled synthetic evaluation only. It does not establish real-world de-anonymization performance.

Decision-Support Notice: This report does not establish real-world identity or guilt.